Code No: 136AG **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Name the elements of an agent.	[2]
b)	Define Artificial Intelligence (AI).	[3]
c)	What do you mean by semantic network?	[2]
d)	Define predicate logic.	[3]
e)	What is an expert system?	[2]
f)	What is the future scope of expert systems?	[3]
g)	Define Reinforcement Learning	[2]
h)	What do you mean by inductive learning?	[3]
i)	What is the importance of the Natural Language Processing?	[2]
j)	Write the significance of parsing in natural language processing.	[3]

PART – B

(50 Marks)

2.a)	Explain the control strategies used to prepare production system.	
b)	What is Alpha-Beta pruning? Explain with the help of suitable example.	[5+5]
	OR	
3.	Discuss in detail heuristic search techniques with example.	[10]
4.a)	Determine whether the following is Satisfiable, Contradictory or Valid: *(P V Q) ^ ~ (P V Q)*	
b)	What are the properties of a good system for the knowledge representation? Discuss.	[5+5]
	OR	
5.	List and explain the various approaches of knowledge representation.	[10]
6.a)	Describe in detail various phases in building Expert systems.	
b)	List and explain various capabilities of the Expert System.	[5+5]
	OR	
7.	Write a detailed note on Bayesian Belief Networks.	[10]

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8.a) List and explain the various design issues of Artificial Neural Networks.
b) Discuss the process of inductive learning using decision trees. [5+5]

OR

9.a) Explain in detail about Single-Layer Feed-Forward Networks with a neat sketch.
b) Write a detailed note on Support Vector Machines. [5+5]

10.a) List various components of natural language understanding process.
b) Describe Syntactic analysis and Semantic analysis in brief. [5+5]

OR

11.a) Write a detailed note on Universal Networking Knowledge.
b) Discuss the steps involved in natural language processing with an example. [5+5]

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